Lesson 23: Implementing a Proportional (P) Controller on a VEX IQ Robot

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Objective:

In this lesson, students will learn about Proportional (P) controllers and apply them to control their VEX IQ robot’s movements accurately. By the end of the lesson, students should understand the basics of Proportional control, tune the `Kp` parameter, and implement a simple Proportional control algorithm to improve the robot's ability to reach a target position with accuracy.

Lesson Outline (2 Hours)

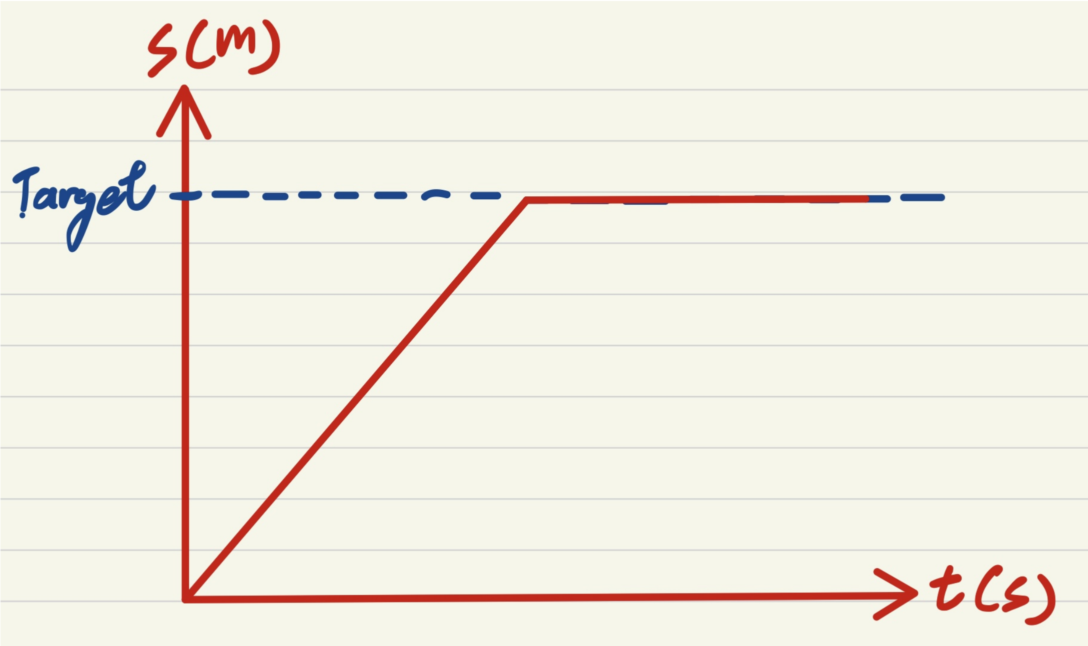
- **Part 1: Introduction to Proportional Controllers and Basic Theory**
- **Part 2: Programming the Proportional Controller**
- **Part 3: Testing and Tuning the Proportional Controller**

Materials Needed:

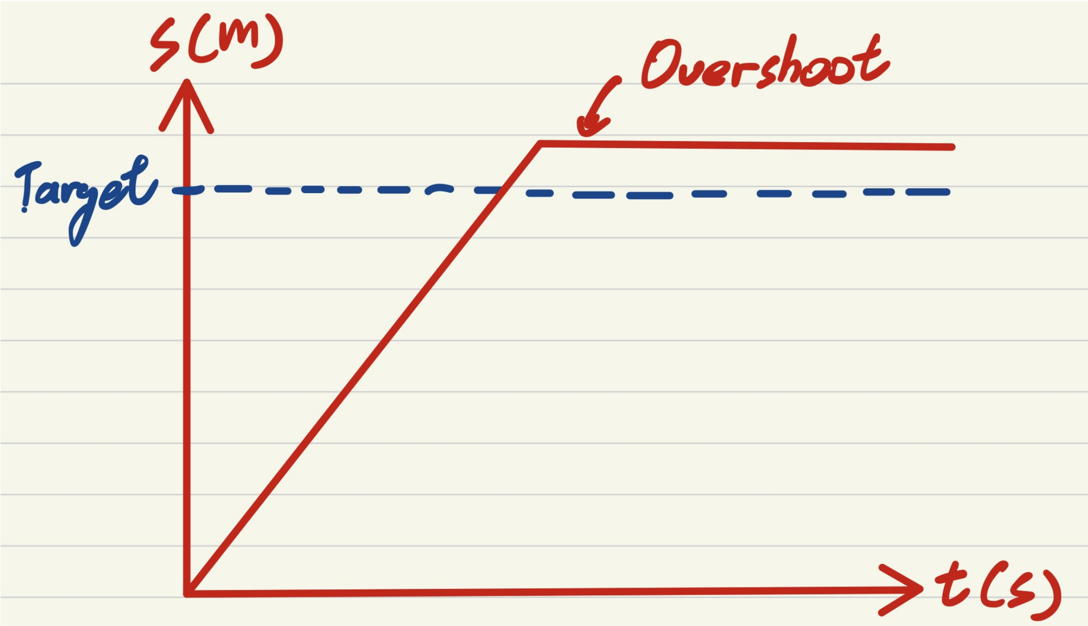
- VEX IQ Robot with Drivetrain and Gyro Sensor
    - If using a Gen 2 brain, then there is no need to use the gyro sensor
- Computers or tablets with VEXcode IQ software

Part 1: Introduction to Proportional Controllers and Basic Theory

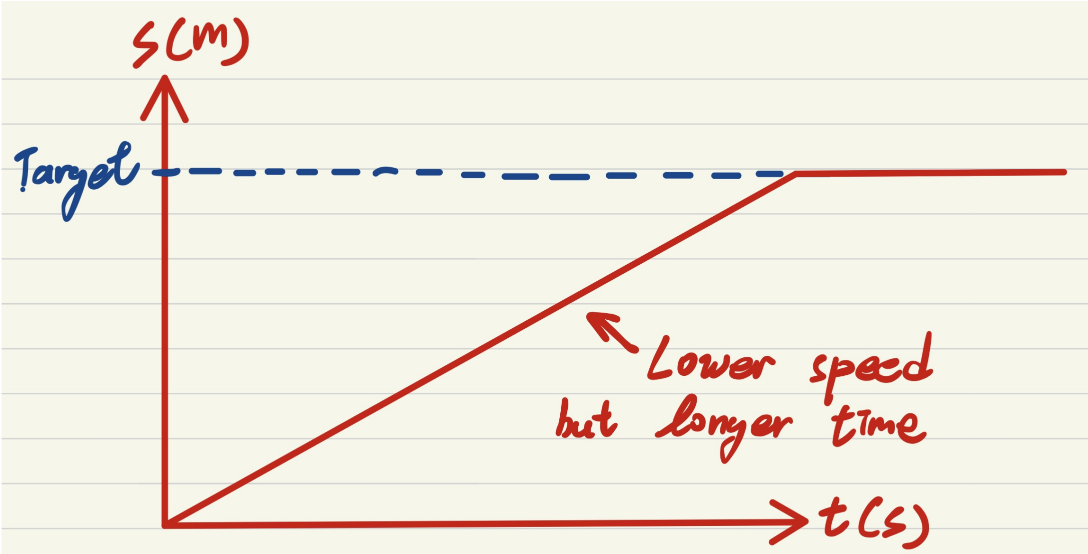
1. **Introduction to Control Systems:**
    - Explain what the “P” controller is and how it automates robotic movement.



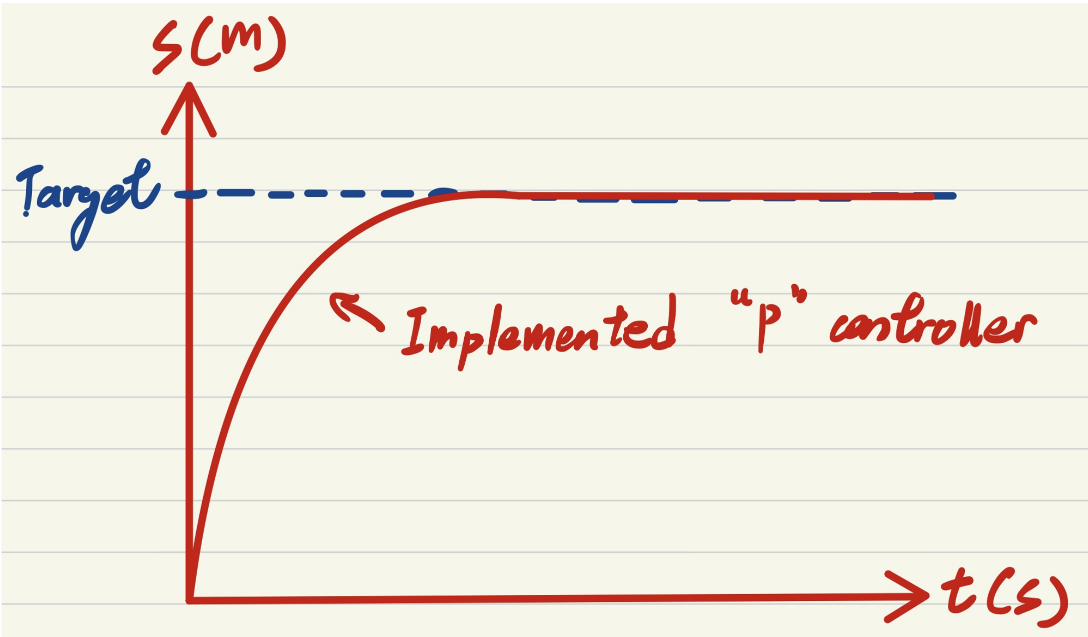
The ideal situation of the robot movement



However, in the real world situation, the overshoot problem will occur.

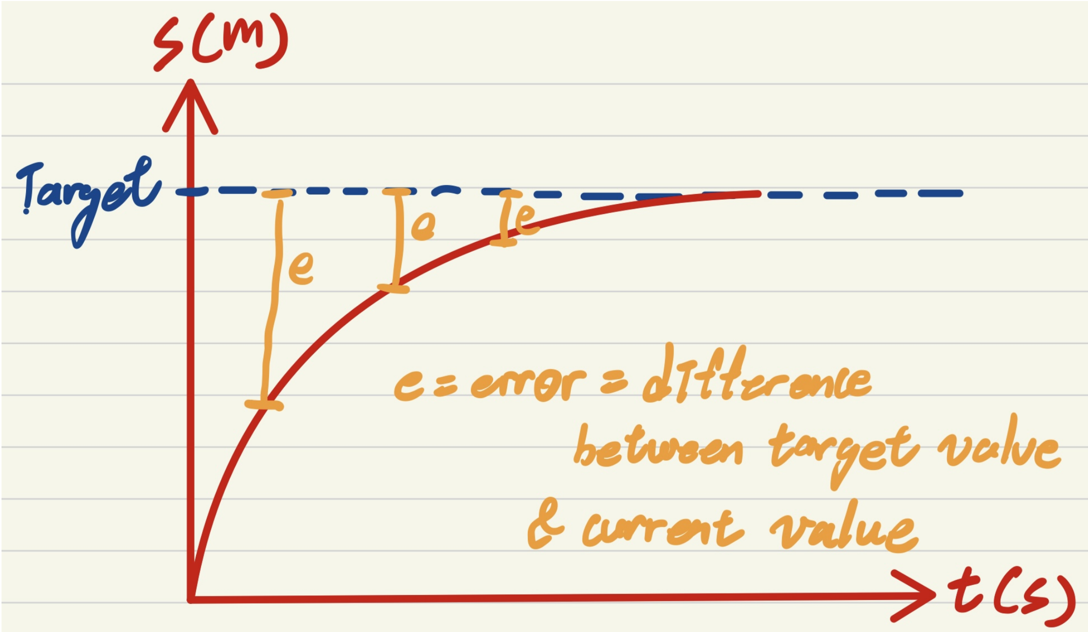


To reduce the overshoot problem, we may lower down the speed. However, this may cause us more time to reach the target.



The use to "P" controller is to slower down the speed gradually which minimise the overshoot problem.

- Discuss the concept of error, which is the difference between the target position and the actual position of the robot.

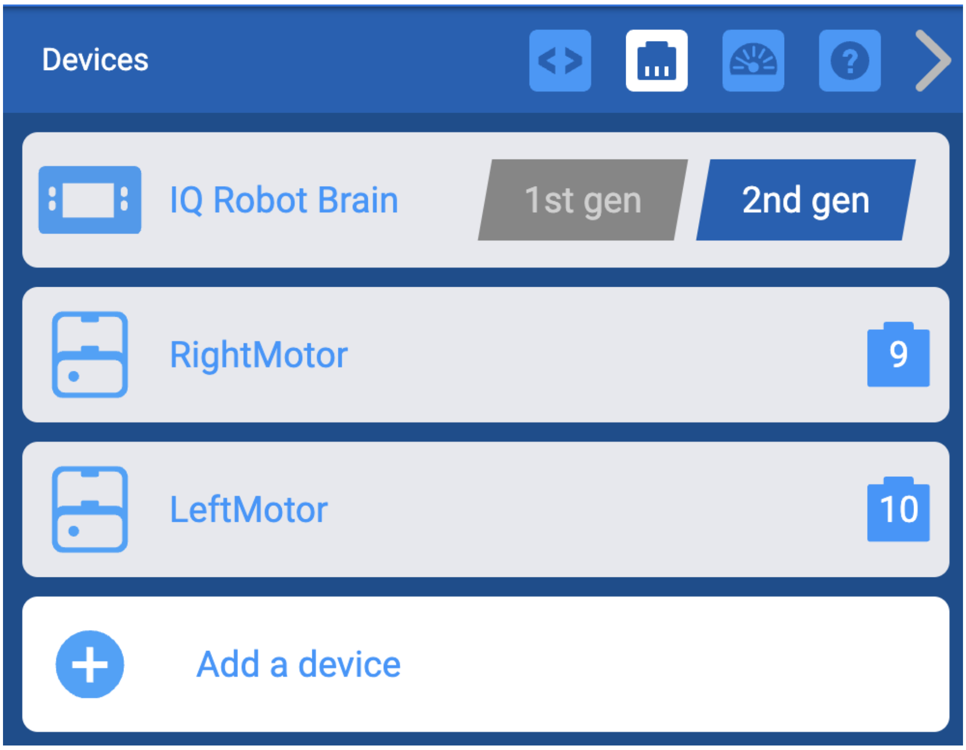


The “P” controller is responsible to the error.

2. **Understanding Proportional Control:**
   - Explain the Proportional controller:
     - The controller applies a correction force proportional to the size of the error.
     - The larger the error, the larger the adjustment.
   - Discuss how the `Kp` value (Proportional Gain) affects the system:
     - A high `Kp` will make the robot correct errors faster but may cause overshooting.
     - A low `Kp` will make the robot correct errors slower but more steadily.
3. **Proportional Formula:**
   - Present the Proportional formula: `Output = Kp * Error`, where:
     - **Kp** is the proportional gain.
     - **Error** is the difference between the target and the actual position.
4. **Setting Up the Objective:**
   - Explain that the goal is to use a Proportional controller to make the robot move to a specific position or maintain a straight path.

Part 2: Programming the Proportional Controller

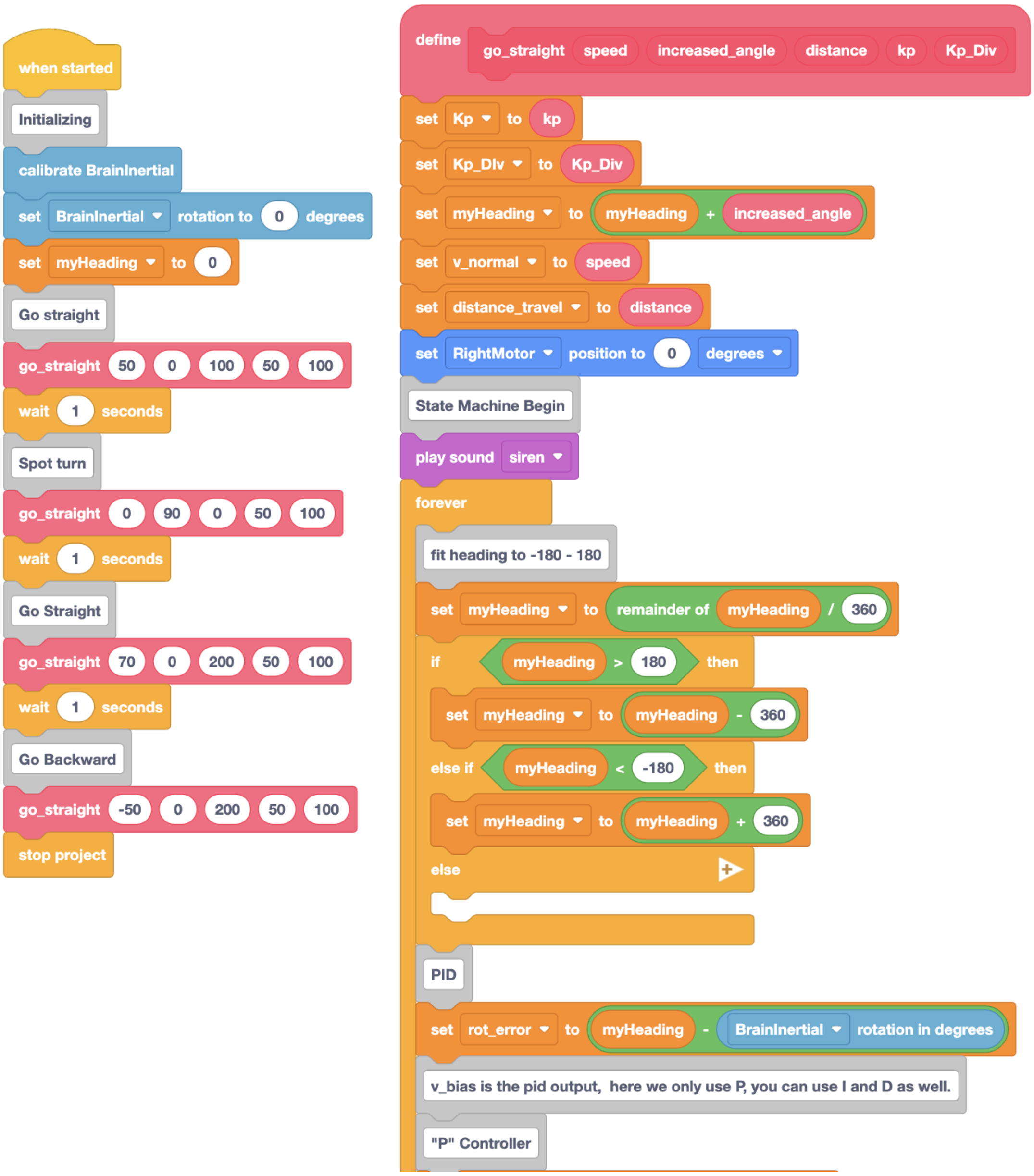
1. **Setup in VEXcode IQ:**
   - Open VEXcode IQ, create a new project, and configure the drivetrain and sensors (e.g., Gyro Sensor or Distance Sensor) as needed.

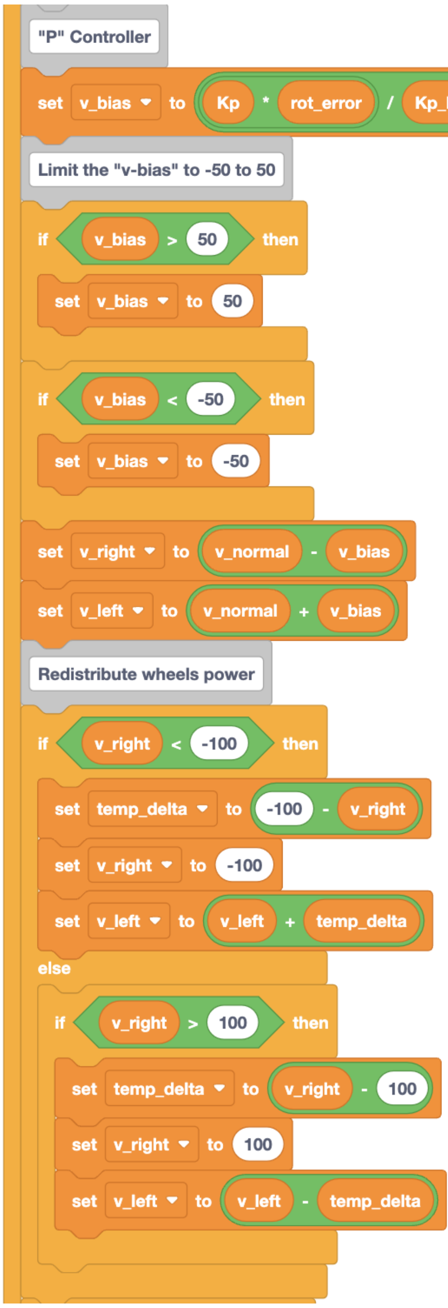


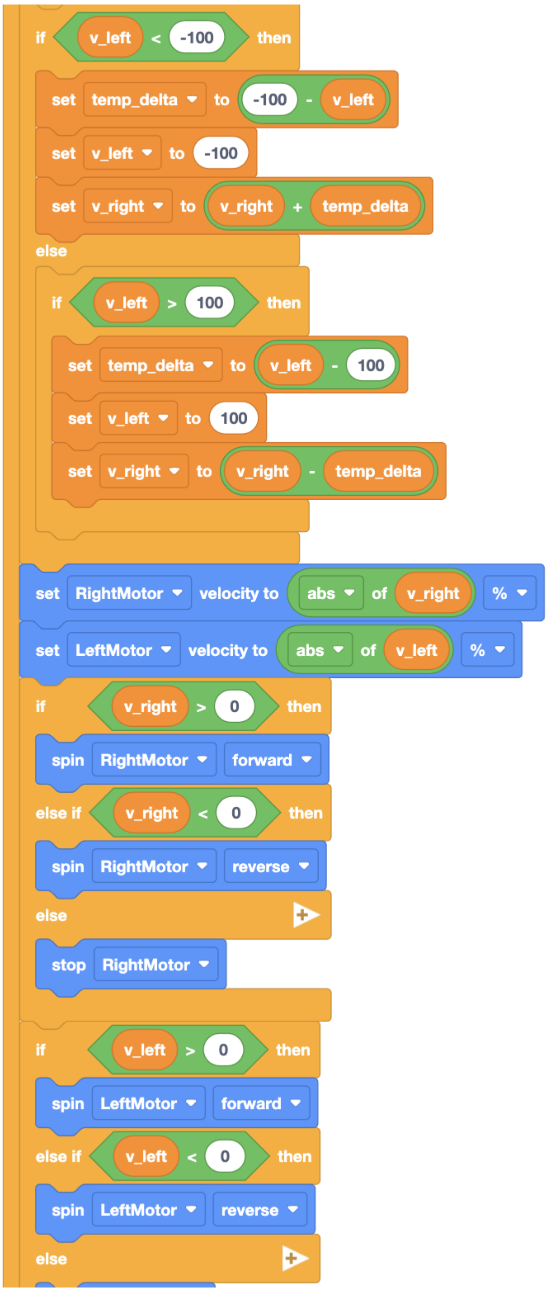
Configuration of the Vexcode IQ

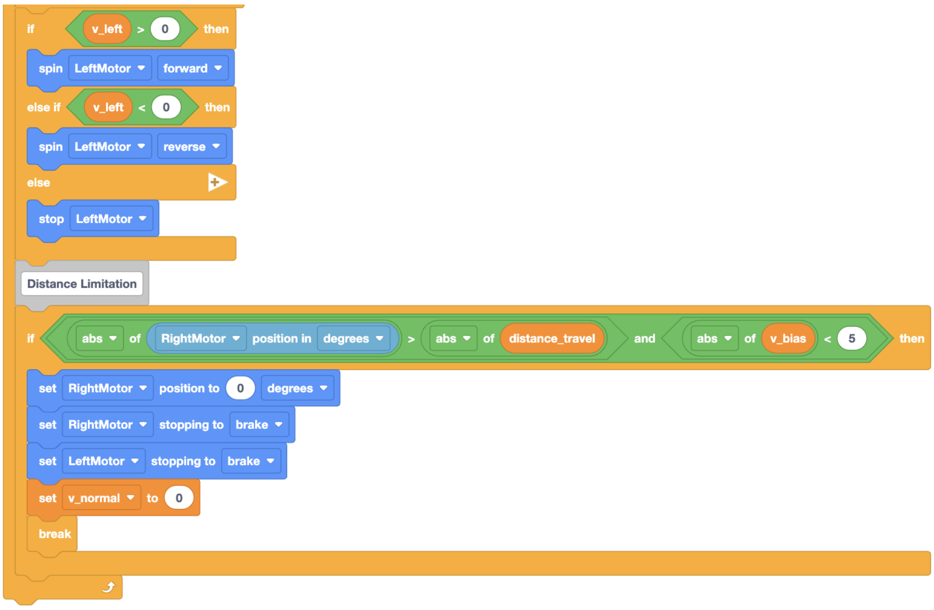
2. **Define Variables for Proportional Components:**
   - Create variables for `Kp` (proportional gain) and `Error`.
   - Explain that students will manually adjust `Kp` to optimize the robot's performance.
3. **Calculate Error in a Loop:**
   - In the main loop, calculate the `Error` as the difference between the target and current position (e.g., using the Gyro or Distance Sensor values).
4. **Implement Proportional Control Logic:**
   - Multiply the `Error` by `Kp` to determine the `Output`.
   - Use the `Output` to adjust the motor speed, making the robot move towards the target position.

5. Code Example:




"P" Controller
set v_bias to Kp * rot_error / Kp_DIv
Limit the "v-bias" to -50 to 50
if v_bias > 50 then
set v_bias to 50
if v_bias < -50 then
set v_bias to -50
set v_right to v_normal - v_bias
set v_left to v_normal + v_bias
Redistribute wheels power
if v_right < -100 then
set temp_delta to -100 - v_right
set v_right to -100
set v_left to v_left + temp_delta
else
if v_right > 100 then
set temp_delta to v_right - 100
set v_right to 100
set v_left to v_left - temp_delta


if v_left < -100 then
set temp_delta to -100 - v_left
set v_left to -100
set v_right to v_right + temp_delta
else
if v_left > 100 then
set temp_delta to v_left - 100
set v_left to 100
set v_right to v_right - temp_delta
set RightMotor velocity to abs of v_right %
set LeftMotor velocity to abs of v_left %
if v_right > 0 then
spin RightMotor forward
else if v_right < 0 then
spin RightMotor reverse
else
stop RightMotor
if v_left > 0 then
spin LeftMotor forward
else if v_left < 0 then
spin LeftMotor reverse
else



6. **Run Initial Test:**
   - Run the program and observe how the robot moves. Record any overshooting or slow response.

Part 3: Testing and Tuning the Proportional Controller

1. **Initial Observations:**
   - Observe how the robot behaves with the initial `Kp` setting. Note if it is moving too quickly, too slowly, or oscillating around the target.
2. **Tuning Strategy:**
   - **Step 1**: Adjust `Kp`:
     - If the robot overshoots the target, reduce `Kp`.
     - If the robot responds too slowly, increase `Kp`.
   - Emphasize that students should find a balance to make the robot move accurately to the target without excessive oscillation or delay.
3. **Testing with Adjustments:**
   - After each adjustment, run the program and observe the effect of the new `Kp` value. Note how the robot's movement changes.
   - **Example Goal**: Adjust `Kp` until the robot reaches the target position smoothly and stops without oscillating.

4. **Wrap-Up Discussion:**
    - Discuss how students tuned the `Kp` parameter and the impact it had on the robot's accuracy and stability.
    - Reinforce the concept that Proportional control alone may not eliminate all steady-state errors, but it's a foundational concept in control systems.

Lesson Wrap-Up

- **Review Key Concepts**: Basics of Proportional control, impact of `Kp` on robot behavior, and limitations of using only a P controller.
- **Q&A**: Address any questions students may have about the Proportional controller concept or implementation.
- **Real-World Applications**: Discuss how Proportional control is used in simple systems like basic speed regulation, temperature control, and feedback-based adjustments in robotics.

This lesson gives students a practical introduction to control systems with a focus on Proportional control. They'll experience firsthand how adjusting control parameters affects the robot's performance, providing a solid foundation for more advanced control strategies in future lessons.